



DEVELOPMENT AND ANALYSIS OF EPOXY -Fe₃O₄ -CARBON COMPOSITE FOR RADAR STEALTH APPLICATIONS

END SEMESTER POSTER PRESENTATION

Submitted by : Ayush Saw (2023meb1331) , Baljinder Kaur (2023meb1334)
Jasnoor Kaur (2023meb1351) , Kondala Seershika (2023meb1355)

Supervisor : Prof.Harpreet Singh

Co -Supervisor : Dr.Satwinder Jit Singh



INTRODUCTION

Stealth technology reduces the radar detectability of objects by minimizing reflected electromagnetic waves. Radar Absorbing Materials (RAMs) absorb incident waves through dielectric and magnetic losses, impedance matching, and multiple internal reflections.

In this project, an epoxy-Fe₃O₄-carbon composite coating was developed for microwave absorption and stealth applications in aircraft, UAVs, military vehicles, and electromagnetic shielding systems.

PROBLEM STATEMENT

- 1.Pure aluminium surfaces strongly reflect incident electromagnetic waves, resulting in poor microwave absorption performance.
- 2.Achieving effective impedance matching and electromagnetic attenuation using lightweight coatings remains a major challenge in stealth applications.
- 3.Therefore, epoxy-Fe₃O₄-carbon composite coatings were investigated to improve microwave absorption and reduce reflection loss (S11).

METHODOLOGY

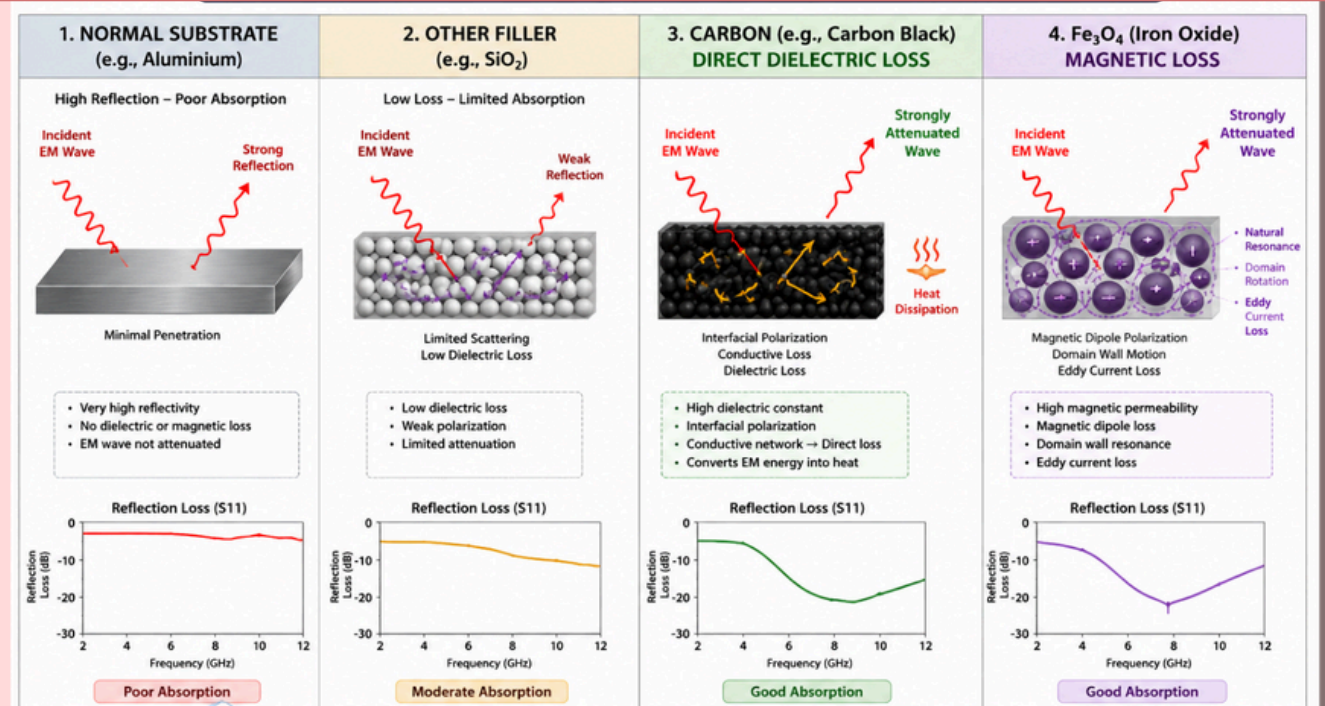


SAMPLE COMPOSITIONS

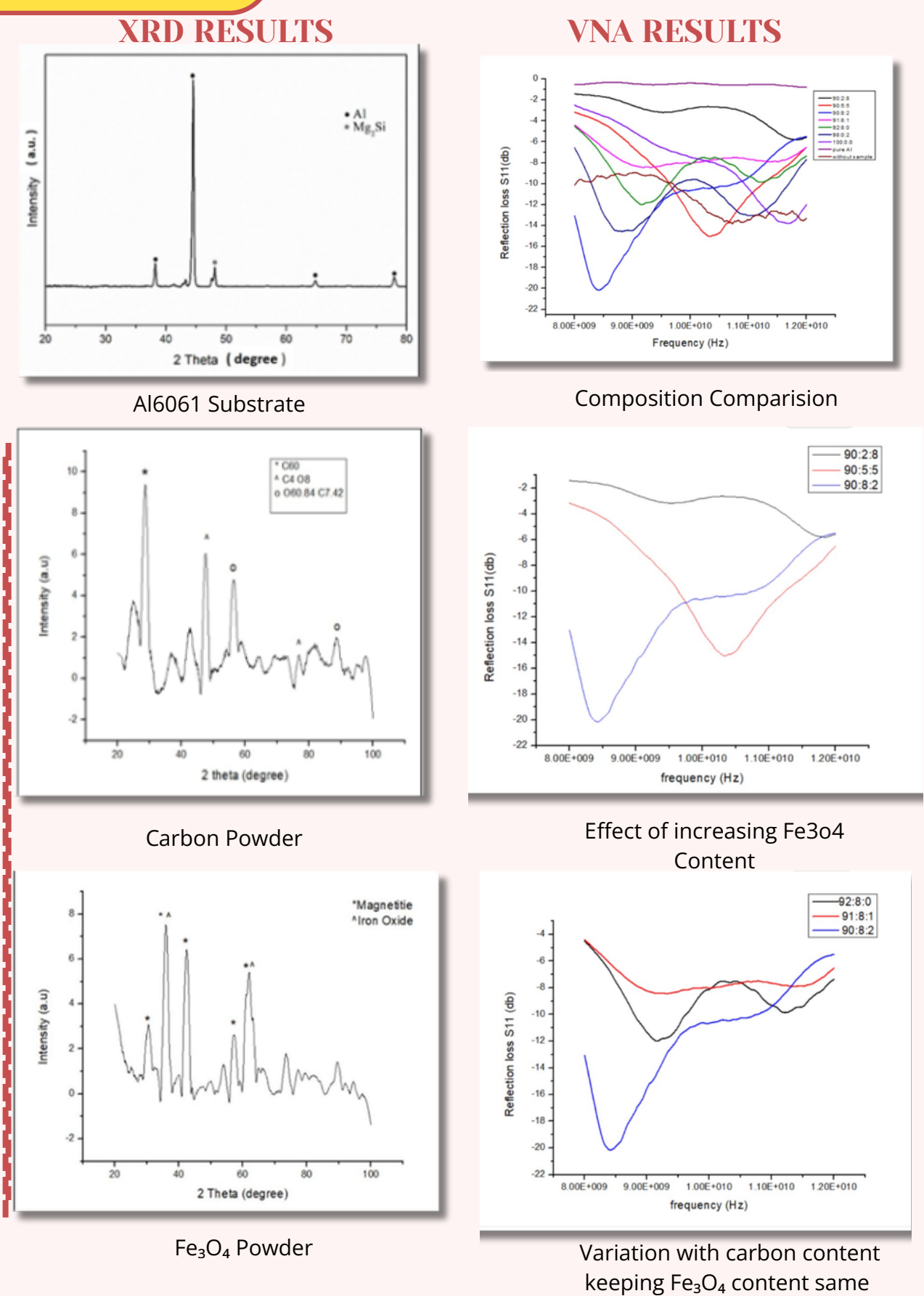
Sample	Epoxy + Hardener	Fe ₃ O ₄ (%)	Carbon (%)
S1	90	8	2
S2	90	5	5
S3	90	2	8
S4	91	8	1
S5	92	8	0
S6	98	0	2
S7	100	0	0

ACKNOWLEDGEMENT

We sincerely acknowledge Prof. Harpreet Singh and Dr. Satwinder Jit Singh for allowing us to undertake this project. I also thank Dr. Lopamudra Palodhi Ma'am, Varun Kumar, and Gaurav Rai from the Surface Engineering Lab for their technical assistance in conducting the deposition and experiments. I am grateful to the staff of the Electrical Department for facilitating the VNA test. I also extend my thanks to the Surface Engineering Lab, Mechanical Engineering Department, for providing access to their facilities.



RESULT



CONCLUSION

- Epoxy-Fe₃O₄-carbon composite stealth coatings were successfully fabricated and analyzed in the X-band frequency range using VNA-based (S11) measurements.
- The addition of Fe₃O₄ and carbon particles significantly improved microwave absorption by enhancing magnetic loss, dielectric loss, and impedance matching.
- Among all compositions, the 90:8:2 (Epoxy:Fe₃O₄:Carbon) sample showed the best absorption performance with reflection loss nearly reaching -20 dB.
- Optimized filler composition provided improved electromagnetic attenuation and more stable absorption behavior compared to pure aluminium and pure epoxy samples.
- The developed lightweight and cost-effective coating shows strong potential for radar stealth, electromagnetic shielding, and future aerospace applications.

REFERENCES

- 1.CB/EP composite absorber study (Liao et al., 2026)
- 2.Introduction to Radar Systems, 3rd ed., McGraw-Hill, 2001.
- 3.Keysight Technologies, Vector Network Analyzer Fundamentals, Application Note, 2022.